

Lecture 15

The Poincaré group and the classification of particles

Objectives. *The finale executes the program. Adding translations to the Lorentz group gives the Poincaré group — a semidirect product in the sense of Lecture 2, realised as 5×5 matrices — and the fields-versus-states principle of Lecture 14 says that its infinite-dimensional unitary irreducible representations are the elementary particles. We classify them by Wigner’s method: diagonalise the translations, watch momenta sweep out orbits — mass shells and the light cone, with their invariant measures derived separately — and reduce everything to the little group of one reference momentum: $SU(2)$ for massive particles, whence spin; the double cover of the Euclidean plane group for massless ones, whence helicity and the photon’s two polarisations, alongside the mathematically allowed continuous-spin branch. For massive particles the Casimirs recover mass and spin; for finite-helicity massless particles both Casimirs vanish and helicity remains a little-group character. A one-page retrospect closes the course by stating its boundary: the representation-theoretic vocabulary needed for $SU(3)$ has been prepared, while particle-physics $SU(3)$ itself is assigned to explicit further reading.*

15.1 The Poincaré group

The interval is not only rotation- and boost-invariant; it is translation-invariant, since it depends on differences of events. The full symmetry group of special relativity is therefore the affine maps

$$x \mapsto \Lambda x + a, \quad \Lambda \in SO^+(1, 3), \quad a \in \mathbb{R}^4,$$

composing as

$$(a, \Lambda) (a', \Lambda') = (a + \Lambda a', \Lambda \Lambda') : \quad (15.1)$$

the *restricted Poincaré group* $\mathcal{P}^+ = \mathbb{R}^4 \rtimes SO^+(1, 3)$, Lecture 2’s semidirect product with continuous ingredients — the translations a normal subgroup, the Lorentz group acting on them by its defining representation. It is a matrix group by the affine trick: let (a, Λ) act on $(x, 1)^T \in \mathbb{R}^5$ as

$$g(a, \Lambda) = \begin{pmatrix} \Lambda & a \\ 0 & 1 \end{pmatrix},$$

a closed subgroup of $GL(5, \mathbb{R})$; block multiplication reproduces (15.1). Differentiating, the Lie algebra consists of $\begin{pmatrix} W & b \\ 0 & 0 \end{pmatrix}$ with $W \in \mathfrak{so}(1, 3)$, $b \in \mathbb{R}^4$: ten dimensions, with four new abstract generators p_μ (translation along e_μ) joining L_a and K_a . The new brackets are two block multiplications:

$$[p_\mu, p_\nu] = 0, \quad [W, p_b] = p_{Wb} \quad \text{for } W \in \mathfrak{so}(1, 3), \quad (15.2)$$

the second saying that the four translations transform as a four-vector; spelled out on the basis,

$$[L_a, p_0] = 0, \quad [L_a, p_b] = \epsilon_{abc} p_c, \quad [K_a, p_0] = p_a, \quad [K_a, p_b] = \delta_{ab} p_0.$$

The physics enters through Lecture 9's dictionary, now once and for all. For an honest strongly continuous unitary representation U with derived representation π , define the self-adjoint translation observables

$$\widehat{P}_\mu := i\pi(p_\mu), \quad U(te_\mu) = e^{t\pi(p_\mu)} = e^{-it\widehat{P}_\mu}. \quad (15.3)$$

Thus p_μ is an abstract Lie-algebra element, $\pi(p_\mu)$ is its generally unbounded skew-adjoint derived operator, and $\widehat{P}_\mu$ is the self-adjoint physical energy-momentum observable. With $\widehat{J}_a = i\pi(L_a)$ and $\widehat{K}_a = i\pi(K_a)$, the abstract brackets translate on a common invariant domain — for the explicit representations below, a dense space of smooth compactly supported wave functions on the orbit (in the massive case ordinary V_j -valued functions; in the massless case, where the standard transformations are chosen only measurably, smooth compactly supported sections of the induced bundle) — to

$$[\widehat{P}_\mu, \widehat{P}_\nu] = 0, \quad [\widehat{J}_a, \widehat{P}_0] = 0, \quad [\widehat{J}_a, \widehat{P}_b] = i\epsilon_{abc} \widehat{P}_c, \quad [\widehat{K}_a, \widehat{P}_0] = i\widehat{P}_a, \quad [\widehat{K}_a, \widehat{P}_b] = i\delta_{ab} \widehat{P}_0.$$

Equivalently set $\widehat{M}_{0a} = \widehat{K}_a$ and $\widehat{M}_{ab} = -\epsilon_{abc} \widehat{J}_c$. With $\eta = \text{diag}(+1, -1, -1, -1)$, the covariant observable brackets used below are

$$\begin{aligned} [\widehat{M}_{\alpha\beta}, \widehat{P}_\nu] &= i(\eta_{\alpha\nu} \widehat{P}_\beta - \eta_{\beta\nu} \widehat{P}_\alpha), \\ [\widehat{M}_{\alpha\beta}, \widehat{M}_{\rho\sigma}] &= i(\eta_{\alpha\rho} \widehat{M}_{\beta\sigma} - \eta_{\alpha\sigma} \widehat{M}_{\beta\rho} - \eta_{\beta\rho} \widehat{M}_{\alpha\sigma} + \eta_{\beta\sigma} \widehat{M}_{\alpha\rho}). \end{aligned} \quad (15.4)$$

Stone's theorem supplies the self-adjoint closures. In this notation the four-momentum is a four-vector without collapsing the three levels of the dictionary.

One paragraph of thesis before the work. A relativistic quantum system carries, by Lecture 11's reasoning, a *projective* unitary representation of $\mathcal{P}^+$. The cure is the one rehearsed twice already: pass to the universal cover. Replacing $\text{SO}^+(1, 3)$ by $\text{SL}(2, \mathbb{C})$ (acting on translations through Φ) gives

$$\mathcal{P} := \mathbb{R}^4 \rtimes \text{SL}(2, \mathbb{C}),$$

topologically $\mathbb{R}^4 \times S^3 \times \mathbb{R}^3$ by Lecture 14 — simply connected — and covering $\mathcal{P}^+$ two-to-one. We now use the Poincaré-specific cohomological lifting result (the “Poincaré lifting” row of the syllabus's import map): for the connected group $\mathcal{P}^+$, the Poincaré Lie algebra satisfies $H^2(\mathfrak{p}, \mathbb{R}) = 0$, and every continuous projective unitary representation pulls back to an honest strongly continuous unitary representation of the simply connected cover $\mathcal{P}$, inducing the same action on rays. This is not a consequence of semisimplicity — $\mathfrak{p}$ is not semisimple — and is distinct from Lecture 14's Lorentz-factor lifting. Thus, unlike the Galilei case with its irremovable mass phase, no central charge survives here. Wigner's definition closes the circle: an *elementary particle* is a system whose state space carries an irreducible such representation — elementary because irreducible: no invariant subspace, no way to take it apart. Lecture 14 proved these representations must be infinite-dimensional; Lecture 13 taught us not to flinch. The rest of the lecture classifies them.

Scope signal. The ray action is projective; the lifting import replaces it by an honest strongly continuous unitary representation of the cover on a separable Hilbert space. Its derived operators are unbounded and share only stated invariant domains. This is a different category from Lecture 14's finite-dimensional nonunitary field representations, and finite character, trace, and dimension theorems do not transfer automatically.

15.2 Momentum and the orbit picture

Wigner’s method begins where the structure suggests: with the abelian normal subgroup. The irreducible unitary representations of $\mathbb{R}^4$ are (by the SNAG theorem below, which packages the Hilbert-space form of Schur’s lemma) its characters $a \mapsto e^{-i\eta(p,a)}$, labelled by points p of momentum space — the plane waves: Lecture 12’s “Fourier series is character theory” for the circle, four noncompact times over. Diagonalising the commuting unitaries $U(a)$ simultaneously is the imported SNAG theorem (Stone–Naimark–Ambrose–Godement): a strongly continuous unitary representation of the locally compact abelian group $\mathbb{R}^4$ on a separable Hilbert space has a unique projection-valued measure on its dual such that $U(a) = \int e^{-i\eta(p,a)} dE(p)$. This is a spectral integral, not a finite-dimensional simultaneous eigenbasis (Berndt, Theorem 5.2, p. 71; Barut–Rączka, Ch. 5). It therefore organises states by momentum, and the Lorentz half of the group shuffles the labels: from (15.1),

$$U(A)U(a)U(A)^{-1} = U(\Lambda a), \quad \Lambda = \Phi(A),$$

so if a state has momentum p then $U(A)$ carries it to one of momentum Λp — the computation is one line with $\eta(p, \Lambda^{-1}a) = \eta(\Lambda p, a)$. In an *irreducible* representation the momenta in play must therefore form a single $\text{SO}^+(1, 3)$ -orbit (heuristically clear; honestly, part of the imported theorem below). The orbits can be listed completely.

Proposition 15.1 (The orbit catalogue). *The orbits of $\text{SO}^+(1, 3)$ on $\mathbb{R}^4$ are: the mass shells $X_m^\pm = \{\eta(p, p) = m^2, \pm p^0 > 0\}$ for each $m > 0$; the two halves of the light cone $X_0^\pm = \{\eta(p, p) = 0, p \neq 0, \pm p^0 > 0\}$; the origin $\{0\}$; and the one-sheeted hyperboloids $\{\eta(p, p) = -\mu^2\}$, $\mu > 0$.*

Figure 15.1 draws the catalogue and, looking ahead to the next section, the little group each orbit carries: the forward mass shell with its $\text{SU}(2)$, the forward cone with $\tilde{\text{E}}(2)$, the excluded spacelike sheet, and the vacuum at the origin.

Proof. The quantity $\eta(p, p)$ is invariant, and for $\eta(p, p) \geq 0$, $p \neq 0$, so is $\text{sgn } p^0$: there $|\vec{p}| \leq p^0$ or $\leq -p^0$, and the same Cauchy–Schwarz estimate that separated the components in Lecture 14 gives $|\sum_i \Lambda^0_i p^i| < \Lambda^0_0 |p^0|$, so $(\Lambda p)^0$ keeps the sign of p^0 — orthochronicity protecting causality. Conversely each listed set is one orbit: a timelike p with $p^0 > 0$ is boosted to rest, $(m, \vec{0})$, by the pure boost along $\vec{p}$ with $\tanh \chi = |\vec{p}|/p^0$; a null p with $p^0 > 0$ is rotated to point along e_3 and then boosted along e_3 — which by Lecture 14’s boost algebra scales $(E, 0, 0, E) \mapsto (e^\chi E, 0, 0, e^\chi E)$ — onto any reference point of the upper cone; spacelike vectors are handled the same way. ■

Physics has named everything in the catalogue: X_m^+ is the *mass shell* of a particle of mass m , and energy $p^0 = E_p = \sqrt{m^2 + |\vec{p}|^2} \geq m > 0$ being bounded below is exactly the choice of the upper sheet (the lower sheet carries negative-energy representations, excluded by the spectrum condition; antiparticles are *positive*-energy states on a forward shell with conjugate charges — the field-theoretic reinterpretation of negative-frequency modes is a different story); the origin will be the vacuum; the spacelike orbits wait, ominously, for the end of the lecture. To do analysis on an orbit we need its invariant measure, and the shell trick of Lecture 12 delivers it one signature up.

Proposition 15.2 (The invariant measure on the mass shell). *For continuous, compactly supported f define*

$$\int_{X_m^+} f d\mu := \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_{m^2 \leq \eta(p,p) \leq m^2 + \varepsilon, p^0 > 0} f(p) d^4 p = \int_{\mathbb{R}^3} f(E_p, \vec{p}) \frac{d^3 p}{2E_p}. \quad (15.5)$$

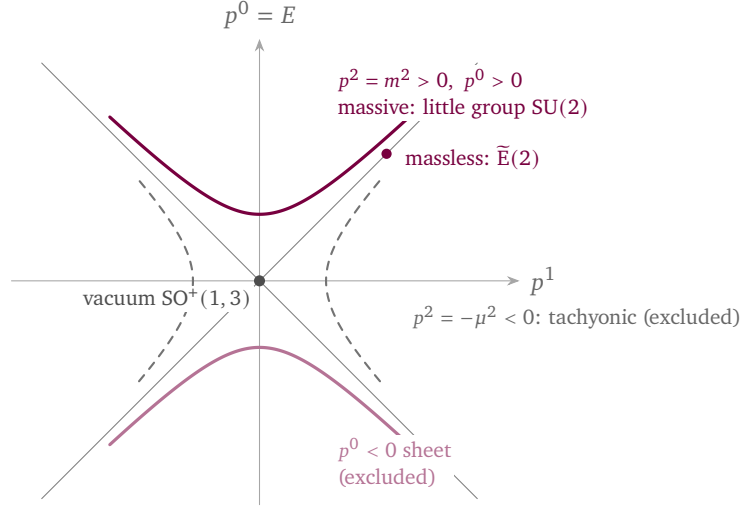


Figure 15.1: The orbits of the Lorentz group on momentum space, and the Wigner classification they carry. Lorentz transformations preserve $p^2 = (p^0)^2 - |\vec{p}|^2$ and the sign of p^0 , so they move momenta only along the level sets drawn here. The physical orbits are the forward mass hyperboloid $p^2 = m^2 > 0$, $p^0 > 0$, whose little group (the stabiliser of a rest momentum) is $SU(2)$, the cover of the geometric rotation group $SO(3)$ — its irreducible representations are the spins; and the forward light cone $p^2 = 0$, whose little group is the double cover $\tilde{E}(2)$ of the Euclidean group and whose finite-helicity representations are labelled by helicity, with a separate continuous-spin branch. The lower sheet $p^0 < 0$ is excluded by positivity of energy; the spacelike orbit $p^2 < 0$ (tachyonic) is excluded, and the origin $p = 0$ is fixed by all of $SO^+(1, 3)$ — the vacuum. A relativistic particle is an orbit together with an irreducible representation of its little group.

This measure is $SO^+(1, 3)$ -invariant: $\int f(\Lambda^{-1}p) d\mu(p) = \int f d\mu$.

Proof. Invariance is manifest in the first expression: d^4p is preserved ($|\det \Lambda| = 1$) and so is the thickened shell. For the second, integrate over p^0 at fixed $\vec{p}$: the shell condition $m^2 \leq (p^0)^2 - |\vec{p}|^2 \leq m^2 + \varepsilon$ cuts out an interval of length $\varepsilon/2E_p + O(\varepsilon^2)$ around $p^0 = E_p$, and the limit collapses onto it. ■

The mysterious $2E_p$ under every integral of relativistic quantum theory is therefore not a convention but a Jacobian: the price of an invariant measure, paid once and for all.

The cone requires its own endpoint check; the preceding proposition assumed $m > 0$.

Proposition 15.3 (The invariant measure on the forward light cone). *On the punctured forward cone $X_0^+ = \{p \neq 0 : p^2 = 0, p^0 > 0\}$, the invariant measure is*

$$d\mu_0(p) = \frac{d^3p}{2|\vec{p}|}, \quad \int_{X_0^+} f d\mu_0 = \int_{\mathbb{R}^3 \setminus \{0\}} f(|\vec{p}|, \vec{p}) \frac{d^3p}{2|\vec{p}|}. \quad (15.6)$$

Proof. Start from the Lorentz-invariant distribution $d^4p \theta(p^0) \delta(p^2)$ — θ the step function and δ the Dirac delta, used only through the one-variable rule $\int \delta(g(p^0)) F dp^0 = F/|g'|$ at a simple root of g ; equivalently, thicken the shell as in Proposition 15.2 and take the limit.

For fixed $\vec{p} \neq 0$ its positive-energy root is $p^0 = |\vec{p}|$, and

$$\left\| \frac{\partial((p^0)^2 - |\vec{p}|^2)}{\partial p^0} \right\|_{p^0=|\vec{p}|} = 2|\vec{p}|.$$

Thus integrating first in p^0 gives exactly (15.6), including the factor 2. The vertex is omitted because the simple-root formula degenerates there; it is a null set, and the density is locally integrable since in spherical coordinates $d^3p/(2|\vec{p}|) = \frac{1}{2}r dr d\Omega$. Finally d^4p , $\delta(p^2)$, and the positive sheet are preserved by $SO^+(1, 3)$, proving invariance. This is the massless norm measure used below. ■

Sheet 15, Exercise 4 checks both (15.5) and (15.6) directly under a boost in the z direction.

15.3 Little groups

The orbit tells us where the momenta live; the remaining freedom sits over a single point of it.

Definition 15.4 (Little group). Fix a reference momentum $\hat{p}$ on an orbit. The *little group* is its stabiliser in the cover, $G_{\hat{p}} = \{A \in SL(2, \mathbb{C}) : \Phi(A)\hat{p} = \hat{p}\}$.

Two computations, both two lines in Lecture 14's Hermitian model $X(p) = p^0\mathbb{K} + \vec{p} \cdot \vec{\sigma}$.

Proposition 15.5 (The massive little group). For $\hat{p} = (m, 0, 0, 0)$, $m > 0$: $X(\hat{p}) = m\mathbb{K}$, and $A(m\mathbb{K})A^\dagger = m\mathbb{K}$ says exactly $AA^\dagger = \mathbb{K}$. So $G_{\hat{p}} = SU(2)$.

The little group of a massive particle is the rotation group of its rest frame — doubly covered, as quantum mechanics insists. Lecture 10's classification is about to be promoted from theorem to definition.

Proposition 15.6 (The massless little group). For $\hat{p} = (1, 0, 0, 1)$: $X(\hat{p}) = \mathbb{K} + \sigma_3 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, and $AX(\hat{p})A^\dagger = X(\hat{p})$ forces the first column of A to be $(\alpha, 0)$ with $|\alpha| = 1$; the determinant then fills in the rest:

$$G_{\hat{p}} = \left\{ \begin{pmatrix} \alpha & \beta \\ 0 & \bar{\alpha} \end{pmatrix} : |\alpha| = 1, \beta \in \mathbb{C} \right\} \cong \mathbb{C} \rtimes U(1),$$

with $\text{diag}(\alpha, \bar{\alpha})$ acting on the normal subgroup $\{\beta\}$ by $\beta \mapsto \alpha^2\beta$. Since $\Phi(\text{diag}(\alpha, \bar{\alpha}))$ is the rotation by the angle $-\arg \alpha^2$ about the ray's direction (Lecture 12's $u_\theta = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$ maps to the rotation by θ), this is the double cover $\tilde{E}(2)$ of the Euclidean group of the plane: rotations and translations of an abstract $\mathbb{R}^2$, with α and $-\alpha$ covering the same rotation.

Proof. Writing $X(\hat{p}) = 2e_1e_1^\dagger$, the condition says $(Ae_1)(Ae_1)^\dagger = e_1e_1^\dagger$, i.e. $Ae_1 = \alpha e_1$ with $|\alpha| = 1$; then $\det A = 1$ forces the diagonal $(\alpha, \bar{\alpha})$ with a free upper entry β . The semidirect structure is the matrix product $\text{diag}(\alpha, \bar{\alpha}) \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \text{diag}(\bar{\alpha}, \alpha) = \begin{pmatrix} 1 & \alpha^2\beta \\ 0 & 1 \end{pmatrix}$. ■

The symmetry group of a flat plane, stabilising a ray of light: an unexpected guest, whose translations will pose a physical riddle in Section 15.5. The general machine that assembles a representation of $\mathcal{P}$ from an orbit and a little-group representation is the course's final structural import (the massless little-group classification, Proposition 15.10, is one further, smaller one).

Theorem 15.7 (Mackey's regular semidirect-product theorem; imported). *Let $N \rtimes H$ be a second-countable locally compact group (a topological group with a countable base of open sets; every matrix Lie group qualifies) with N abelian and with regular H -action on the character group $\widehat{N}$ (a technical condition on the orbits — roughly, that they can be parametrised measurably, as the locally closed momentum orbits of Proposition 15.1 can). Then every irreducible strongly continuous unitary representation on a separable Hilbert space is induced from a character in one H -orbit and an irreducible unitary representation of its stabiliser; the induced representation is irreducible, these exhaust the irreducibles, and equivalence is precisely conjugacy of the orbit-stabiliser data. For $\mathcal{P} = \mathbb{R}^4 \rtimes \mathrm{SL}(2, \mathbb{C})$ these hypotheses hold: the translation subgroup is abelian, both factors are second-countable locally compact Lie groups, and the momentum action has the regular, locally closed orbits of Proposition 15.1. Consequently an irreducible representation is determined, up to unitary equivalence, by a pair: an orbit O of Proposition 15.1 and an irreducible unitary representation σ of the little group of a reference point $\hat{p} \in O$. The representation is realised on $L^2(O, d\mu; V_\sigma)$ — square-integrable V_σ -valued functions on the orbit — by the explicit formulas of the next section (the induced representation).*

Proof boundary and source. SNAG above supplies only the translation projection-valued measure. The orbit reduction, measurable induction, induced irreducibility, exhaustivity, and equivalence clauses are the imported theorem: Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed., Theorem 6.42 and the discussion in §6.6, pp. 199–201. Computing the unitary dual of a particular little group is a further, separate task.

We import the bookkeeping — that the list is exhaustive — and build the entries that matter by hand. The catalogue is now a table: massive orbits carry $\mathrm{SU}(2)$, so their representations are labelled (m, j) with $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ — *mass and spin*; massless orbits carry $\widetilde{\mathrm{E}}(2)$ — *finite helicity* and continuous-spin branches, as we shall see; the origin's little group is all of $\mathrm{SL}(2, \mathbb{C})$, whose only finite-dimensional unitary representation is trivial (Lecture 14's no-go): the trivial pair is the *vacuum*, the unique Poincaré-invariant state, and nature declines the infinite-dimensional alternatives.

15.4 Massive particles: mass and spin

Fix $m > 0$, the orbit X_m^+ with $\hat{p} = (m, 0, 0, 0)$, and a spin $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ with Lecture 10's representation (Π_j, V_j) of the little group $\mathrm{SU}(2)$. The construction needs one choice: a *standard boost* carrying $\hat{p}$ to each $p \in X_m^+$, and Lecture 14's square-root machinery supplies a canonical one. The matrix $X(p)/m$ is positive definite on the upper shell (its eigenvalues are $(p^0 \pm |\vec{p}|)/m > 0$) with determinant one, so

$$S_p := (X(p)/m)^{1/2} \in \mathrm{SL}(2, \mathbb{C}) \quad \text{satisfies} \quad S_p X(\hat{p}) S_p^\dagger = m S_p^2 = X(p) :$$

a positive element — a *pure boost*, by the polar anatomy — with $\Phi(S_p)\hat{p} = p$ and $S_{\hat{p}} = \mathbb{K}$. The whole representation is now three formulas and one telescoping identity.

Theorem 15.8 (The massive representations (m, j)). *On $\mathcal{H} = L^2(X_m^+, d\mu) \otimes V_j$ — wave functions $\psi: X_m^+ \rightarrow V_j$, square-integrable against Proposition 15.2 — define*

$$(U(a)\psi)(p) = e^{-i\eta(p,a)} \psi(p), \quad (U(A)\psi)(p) = \Pi_j(W(A, p)) \psi(\Lambda^{-1}p), \quad (15.7)$$

where $\Lambda = \Phi(A)$ and

$$W(A, p) := S_p^{-1} A S_{\Lambda^{-1}p} \quad (15.8)$$

is the Wigner rotation. Then $W(A, p) \in \text{SU}(2)$, the formulas define a unitary representation $U^{(m,j)}$ of $\mathcal{P}$ (strongly continuous: for ψ continuous with compact support, $U(a, A)\psi \rightarrow \psi$ pointwise as $(a, A) \rightarrow (0, \mathbb{1})$ — the phase, the Wigner rotation and the moving argument $\Lambda^{-1}p$ all vary continuously — with a common compactly supported bound, so in $L^2(d\mu)$ by dominated convergence; density and unitarity extend this to all of L^2). Its irreducibility is the imported irreducibility clause of Theorem 15.7. Distinct pairs (m, j) give inequivalent representations, and by Theorem 15.7 these exhaust the massive orbits.

Proof. $W(A, p)$ stabilises $\hat{p}$ — apply $\Phi: \Phi(S_p)^{-1} \Lambda \Phi(S_{\Lambda^{-1}p})$ sends $\hat{p} \mapsto \Lambda^{-1}p \mapsto p \mapsto \hat{p}$ — so it lies in the little group, which Proposition 15.5 computed to be exactly $\text{SU}(2)$, and Π_j may eat it. The homomorphism property is a telescope:

$$W(A, p) W(A', \Lambda^{-1}p) = S_p^{-1} A (S_{\Lambda^{-1}p} S_{\Lambda^{-1}p}^{-1}) A' S_{(\Lambda\Lambda')^{-1}p} = W(AA', p),$$

whence $U(A)U(A') = U(AA')$ by chasing (15.7); the mixed law $U(A)U(a)U(A)^{-1} = U(\Lambda a)$ is the character identity $\eta(\Lambda^{-1}p, a) = \eta(p, \Lambda a)$; unitarity is Π_j unitary plus the invariance of $d\mu$. Irreducibility and inequivalence now invoke Theorem 15.7. The relevant commutants are measurable matrix-valued multipliers defined only almost everywhere, so evaluating one at the single point $\hat{p}$ is not a valid elementary proof. The required measurable-field/ergodic reasoning is part of the import and is not silently replaced here. ■

Read the formulas physically. At the reference fibre, rotations act by $\Pi_j(U)$ itself — $W(U, \hat{p}) = U$ — so the index of V_j is literally Lecture 10's spin multiplet, sitting in the rest frame: *spin is the little group's quantum number*, and the classification of Lecture 10, proved as a theorem about $\mathfrak{su}(2)$, has been promoted to the definition of what spin is. For a boost A , the Wigner rotation (15.8) is a product of three boosts that is nevertheless a rotation — precisely the rotation debris of Lecture 14's bracket $[K, K] = -\epsilon L$, now with a job description: it is *how a moving particle's spin frame rotates*, the cocycle that Thomas precession was waiting for. Sheet 14's perpendicular-boost angle is exactly $W(A, p)$ for that configuration, and Notebook 15 watches the telescope identity hold to machine precision.

Remark 15.9 (The massive Casimir values). Two observables, symmetric on the common invariant domain (self-adjointness of their closures is not proved here), commute with the generators there. The first is $\hat{P}^2 = \eta^{\mu\nu} \hat{P}_\mu \hat{P}_\nu$, which acts in (15.7) as multiplication by $\eta(p, p) = m^2$. The second — designed to measure the spin part of angular momentum Lorentz-invariantly, as the rest-frame computation below confirms — is built from the Pauli–Lubanski observable

$$\hat{W}^\mu = \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} \{\hat{P}_\nu, \hat{M}_{\rho\sigma}\}, \quad \epsilon^{0123} = 1,$$

on the common invariant domain, where $\hat{M}_{ab} = -\epsilon_{abc} \hat{J}_c$. The anticommutator makes symmetry manifest and equals the usual ordered formula after contraction of the Poincaré brackets. One checks $[\hat{W}^\mu, \hat{P}_\nu] = 0$ and that $\hat{W}^2$ commutes with everything (the full index calculation is carried out in Sheet 15, Exercise 2, the Pauli–Lubanski/Casimir capstone). On the rest fibre the momentum factors freeze to $(m, \vec{0})$, the orbital terms drop out, and $\hat{W} = (0, m\hat{J})$ with $\hat{J}$ the rest-frame rotation generators, so

$$\hat{W}^2 = -m^2 \hat{J}^2 = -m^2 j(j+1) \mathbb{1}.$$

Thus on the massive positive-energy branch the two eigenvalues recover mass and spin. This conclusion is deliberately orbit-specific: on the finite-helicity massless branch both Casimirs vanish and do not recover helicity.

Two footnotes from earlier lectures come due. The wave functions $\psi(p)$ with their $d^3p/2E_p$ norm are the relativistic normalisation of every field-theory textbook, explained: the $2E_p$ buys invariance (Proposition 15.2). And many particles cost no new group-level construction: the state space of n identical such particles is the symmetric or antisymmetric power of $\mathcal{H}$, and every unitary $U^{(m,j)}(g)$ lifts to the everywhere-defined all-Fock unitary $\Gamma(U^{(m,j)}(g))$. A derived generator is a different statement: it is first $d\Gamma(A)$ on the algebraic finite-particle core, is generally unbounded (already the number operator is), and its skew-adjoint closure generates $\Gamma(e^{tA})$. Thus Lecture 13's second-quantisation machine does not make derived all-Fock generators everywhere defined.

15.5 Massless particles: helicity

The massless story differs in exactly one ingredient: the little group is $\widetilde{E}(2)$, and its representation theory has a fork in it.

Proposition 15.10 (Unitary dual of $\widetilde{E}(2)$; Mackey classification imported). *(i) The subgroup $\{\beta\} \cong \mathbb{C}$ acts trivially in finite dimension. Each irreducible is therefore a one-dimensional character of the rotation cover,*

$$\chi_h: \text{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto e^{-ih\theta}, \quad h \in \frac{1}{2}\mathbb{Z},$$

single-valued on the double cover, with $\chi_h(-\mathbb{K}) = (-1)^{2h}$ — rays, one last time.

(ii) A nonzero translation character has a circular orbit of radius $\rho > 0$. Its stabiliser in the rotation cover is $\{\alpha : \alpha^2 = 1\} = \{\pm 1\}$, so for every ρ there are two inequivalent infinite-dimensional irreducibles, distinguished by the two characters of this stabiliser (equivalently, by the central sign). These are the continuous-spin representations. With ρ normalised as the Pauli–Lubanski scale, $\widehat{W}^2 = -\rho^2\mathbb{K}$ in signature $(+ - - -)$.

Proof. The commuting unitaries $U(\beta)$ decompose the space into characters $e^{i\text{Re}(\bar{k}\beta)}$, $k \in \mathbb{C}$, and conjugating by the rotation $\text{diag}(\alpha, \bar{\alpha})$ carries the k -eigenspace to the $\alpha^2 k$ -eigenspace (the semidirect relation of Proposition 15.6, pushed through the representation). The set of occurring labels is therefore invariant under the full circle of rotations; in finite dimensions it is also finite, and the only finite circle-invariant subset of $\mathbb{C}$ is $\{0\}$. With the translations dead, the representation factors through the circle of rotations, whose irreducible representations are the characters listed — the half-integers entering exactly as they did for $\text{SU}(2)$.

For the complete unitary dual, apply the regular semidirect-product theorem separately to $\mathbb{C} \rtimes \text{U}(1)$ with action $\beta \mapsto \alpha^2 \beta$. The dual action sends $k \mapsto \alpha^2 k$. Its orbits are $\{0\}$ and the circles $|k| = \rho > 0$; their stabilisers are respectively $\text{U}(1)$ and $\{\pm 1\}$. The little representations are therefore the characters listed in (i), or one of the two signs in (ii). Folland, Theorem 6.42, supplies induction and irreducibility; Ramond, Ch. 11, §11.1.1, pp. 230–231, records the two cover sectors and the helicity/continuous-spin fork. This little-group classification is not a consequence of merely applying SNAG to the Poincaré translations. ■

The label h is the *helicity*: the eigenvalue of angular momentum projected on the direction of motion, the only rotation a light ray has time for. A massless particle of helicity h is the induced representation over X_0^+ with $\sigma = \chi_h$ — *scalar-valued* wave functions, one component per momentum, with norm

$$\|\psi\|^2 = \int_{\mathbb{R}^3 \setminus \{0\}} |\psi(|\vec{p}|, \vec{p})|^2 \frac{d^3 p}{2|\vec{p}|},$$

using Proposition 15.3, and with a measurable family of standard transformations along the cone (rotation, then boost).

The Casimirs now behave differently. At $\hat{p} = (E, 0, 0, E)$ one has $\hat{P}_0 = E$ and $\hat{P}_3 = -E$. In the finite-helicity branch the two *null-rotation* generators (the derived operators of the translation subgroup $\{\beta\}$ of $\tilde{E}(2)$, which at $\hat{p}$ are the components $\hat{W}^1, \hat{W}^2$ up to the factor E) vanish, while $U(r(\theta)) = e^{-i\theta\hat{J}_3} = e^{-ih\theta}$ gives $\hat{J}_3 = h$. The defining signs therefore give

$$\hat{W}^0 = \frac{1}{2}\epsilon^{03jk}\hat{P}_3\hat{M}_{jk} = E\hat{J}_3, \quad \hat{W}^3 = \frac{1}{2}\epsilon^{30jk}\hat{P}_0\hat{M}_{jk} = E\hat{J}_3,$$

and $\hat{W}^1 = \hat{W}^2 = 0$. Hence

$$\hat{W}^\mu = h\hat{P}^\mu \quad \text{on every finite-helicity massless representation,} \quad \hat{P}^2 = \hat{W}^2 = 0. \quad (15.9)$$

The sign is fixed, not remembered: both displayed nonzero components are $+Eh$, and covariance carries the relation along X_0^+ . Thus the two Casimir eigenvalues do not distinguish the different values of h .

Nature's choice at the fork is empirical and unanimous: every massless particle observed is of type (i). The continuous-spin option — infinitely many polarisation states at each momentum — is a chair at Wigner's table that has never been filled.

Helicity's physics is its behaviour under parity. Spatial inversion reverses $\vec{p}$ but not the axial $\vec{J}$, so it flips $\vec{J} \cdot \hat{p}$: helicity h becomes $-h$ (at group level, conjugation by parity carries the little group of $\hat{p}$ to that of the reversed ray, inverting the rotation character). A parity-symmetric theory must therefore contain $\pm h$ together, and *electromagnetism is the example*: the photon is the pair $h = \pm 1$, and so has exactly *two* polarisation states — where a *massive* spin-one particle has $2j + 1 = 3$. The missing longitudinal photon is a little-group theorem: the third state is a luxury only mass can buy. (The graviton's pair is $h = \pm 2$; the weak interaction, which violates parity, happily employs unpaired helicities.) One last ledger entry: the photon *state* has one component per momentum and helicity, while the photon *field* A_μ has four — the mismatch between $(\frac{1}{2}, \frac{1}{2})$ and $\chi_{\pm 1}$ is real, and the apparatus that manages it is gauge invariance: the final, and deepest, item in the fields-versus-states accounts, payable in a field theory course.

15.6 Retrospect

The positive-energy catalogue is now separated into its branches. An elementary relativistic system is an orbit and a little-group representation: massive particles (m, j) , with spin defined by the rest frame's $SU(2)$; massless particles $(0, h)$, with helicity defined by the ray's $\tilde{E}(2)$ rotation character; the two continuous-spin families at each $\rho > 0$, distinguished by the cover's central sign; and the vacuum, alone on the trivial orbit. In the empirical table below two chairs stand empty: the continuous-spin representations, and the spacelike orbits — whose little group, the stabiliser of $X(\hat{p}) = \mu\sigma_3$ in $SL(2, \mathbb{C})$, is the noncompact group of σ_3 -unitaries $SU(1, 1) = \{A : A^\dagger \sigma_3 A = \sigma_3, \det A = 1\} \cong SL(2, \mathbb{R})$, the double cover of the geometric little group $SO^+(1, 2)$, so that nontrivial unitary little-group representations would give tachyons infinitely many spin states on top of their causal misbehaviour. The trivial unitary representation is the one-dimensional exception. Nature, as far as anyone has looked, declines both.

The dictionary to the particles of nature is then read off a single table — a kinematic one: it records the spin/helicity type of each species' field, and applies literally only to stable, free asymptotic particles (the W , Z and Higgs are unstable; quarks and gluons are confined):

Wigner label	little group	physical example
$(m, 0)$	$SU(2)$	the Higgs boson
$(m, \frac{1}{2})$	$SU(2)$	electron, quarks, massive neutrinos
$(m, 1)$	$SU(2)$	the $W^\pm$ and Z bosons
$(0, \pm 1)$	$\tilde{E}(2)$	the photon, the gluons
$(0, \pm 2)$	$\tilde{E}(2)$	the (conjectured) graviton
origin orbit, trivial irrep	$SL(2, \mathbb{C})$	the vacuum (not $(0, h=0)$, the massless scalar)

The one historical correction sits in the second row: neutrinos were long entered as massless $(0, \pm \frac{1}{2})$, but flavour oscillations prove $m_\nu \neq 0$, moving them onto the massive ladder — the single empirical edit made to Wigner’s 1939 table.

What remains between this classification and physics is one construction: a quantum *field* is a spacetime-indexed family of Fock-space operators intertwining a finite-dimensional (j_1, j_2) — Lecture 14’s fields — with the infinite-dimensional unitary representations built today — Lecture 15’s states. The Dirac equation was the prototype of such an intertwining condition; building them systematically, with locality and causality enforced, is quantum field theory, the natural next course. Its multi-particle arena, Fock space with Γ and $d\Gamma$, left Lecture 13 ready for use.

The course’s $SU(3)$ boundary is deliberate. Duals and conjugate representations, invariant tensors, $SU(2)$ coupling, and highest-weight staircases have prepared the vocabulary and methods. Applying them to the quark $\mathbf{3}$ and antiquark $\bar{\mathbf{3}}$, the eightfold way, roots and weights, and strong-interaction invariant tensors is further study, not material taught in these lectures. Jones, Ch. 8, §§8.3–8.4, pp. 149–164 is the exact next destination. The other horizons are visible from here: symmetry made *local* (gauge theory), symmetry *broken* by a ground state, harmonic analysis and Mackey theory in earnest, and the strange cases where a classical symmetry fails to survive quantisation.

We opened Part I by counting the symmetries of a triangle; we close Part II by defining what a particle is. It was the same calculation throughout — decompose, then intertwine — performed with heavier measures and steadier nerves, and its moral has not changed since the third week: a symmetry is never just a group; it is a group *represented*, and the representation is where the physics lives.

Remark 15.11 (Computer algebra). Notebook 15 draws the orbit catalogue (sheets, cone, hyperboloid); computes standard boosts S_p by matrix square roots and verifies $\Phi(S_p)\hat{p} = p$; checks the Wigner cocycle (15.8) and its telescoping identity on random group elements to machine precision; tracks the helicity phase $e^{-ih\theta}$ around a full 4π loop of rotations; and confirms the invariance of $d^3p/2E_p$ by Monte Carlo against random boosts.

Summary.

- *Poincaré group.* The semidirect product $\mathbb{R}^4 \rtimes SO^+(1, 3)$ of translations and Lorentz transformations (eq. (15.1)).
- *Orbits.* The Lorentz orbits in momentum space are the mass shells, the light cone, the spacelike sheet, and the origin (Proposition 15.1); the massive-shell and punctured-cone measures are derived separately in Propositions 15.2 and 15.3.
- *Little groups.* The stabiliser of a momentum is $SU(2)$ for a massive particle (Proposition 15.5) and $\tilde{E}(2)$ for a massless one (Proposition 15.6).
- *Wigner classification.* An elementary particle is an orbit together with an irreducible representation of its little group (Mackey, Theorem 15.7): massive (m, j) with spin

(Theorem 15.8); massless finite-helicity characters $(0, h)$; and two continuous-spin sectors for each $\rho > 0$.

- *Casimirs*. On the massive branch, $\widehat{P}^2 = m^2$ and $\widehat{W}^2 = -m^2 j(j+1)$ recover mass and spin. On the finite-helicity massless branch $\widehat{W}^\mu = h\widehat{P}^\mu$, so both Casimirs vanish and the rotation character, not the Casimirs, records h .

Tutorial. Exercise Sheet 15.

References

Folland, 2nd ed., §6.6, Theorem 6.42, pp. 199–201 (regular Mackey theorem); Ramond, Ch. 11, §11.1.1, pp. 224–231 (Casimirs and covered massless sectors); Tung, Ch. 10, §10.4, pp. 191–200 (Wigner classification and Pauli–Lubanski relations); Jones, 2nd ed., Ch. 10, §10.4, pp. 208–216 and Appendix E, pp. 279–280 (little groups and on-shell measure). SNAG: Berndt, Theorem 5.2, p. 71; Barut–Rączka, Ch. 5. SU(3) further reading: Jones, Ch. 8, §§8.3–8.4, pp. 149–164.

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